

Creating Platform Library In spring boot

```
access-control-service/pom.xml X
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1<project xmlns="http://maven.apache.org/POM/4.0.0"
2      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://
4
5      <modelVersion>4.0.0</modelVersion>
6
7      <parent>
8          <groupId>org.springframework.boot</groupId>
9          <artifactId>spring-boot-starter-parent</artifactId>
10         <version>4.0.0</version>
11         <relativePath />
12     </parent>
13
14     <groupId>com.pisethjavaschool</groupId>
15     <artifactId>access-control-service</artifactId>
16     <version>1.0.0</version>
17
18     <name>access-control-service</name>
19
20     <properties>
21         <java.version>25</java.version>
22         <spring-cloud.version>2025.1.0</spring-cloud.version>
23         <mapstruct.version>1.6.3</mapstruct.version>
24         <lombok.version>1.18.42</lombok.version>
25     </properties>
26
27     <dependencyManagement>
28         <dependencies>
```

- Normal project in the pom.xml

```
<parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>4.0.0</version>
    <relativePath />
</parent>
```

- What we can get from access-control project to create platform library :

- AuditableEntity (បឹងកន្លែងនឹងវាជា candidate មួយហើយដែលយើងត្រូវធ្វើការ move វាចេញ) កន្លែងនឹងសម្រាប់ធ្វើ Auto
- package com.pisethjavaschool.accesscontrol.common.exception: តាមក្រុមហ៊ុនមួយៗវាមាន កន្លែងដែល handle exception បឹងកន្លែងនឹងយើងត្រូវដាក់វាមួយជុំដែលដើម្បីស្រួលធ្វើការ import ចូល
- package com.pisethjavaschool.accesscontrol.common.security; ផ្នែក security
- package com.pisethjavaschool.accesscontrol.common.web ផ្នែក filter
- package com.pisethjavaschool.accesscontrol.common.openapi; API Swager

របៀបបង្កើត Library បង្កើត multi module project

- platform-common
- platform-exception
- platform-openapi

(edb: Enterprise development)

យើងធ្វើការ extract វាជា Module ហើយបង្កើតវាជា Jar file ដាក់ចូល Project ដោយ add dependency to the project ដែលយើងចង់ហៅវាមកប្រើប្រាស់នឹង

1- Creating parent project

```
pom.xml U X
> Users > home > Desktop > Enterprise Backend > Softcode > ebd-platform-libraries > pom.xml
1  <?xml version="1.0" encoding="UTF-8"?>
2  <project xmlns="http://maven.apache.org/POM/4.0.0"
3      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
5      <modelVersion>4.0.0</modelVersion>
6
7      <groupId>com.pisethjavaschool.platform</groupId>
8      <artifactId>platform-libraries</artifactId>
9      <version>1.0.0</version>
10     <packaging>pom</packaging> ✓
11
12     <name>platform-libraries</name>
13     <description>Reusable platform libraries for the Restaurant and KTV enterprise course.</description>
14
15     <modules>
16         <module>platform-common</module>
17         <module>platform-exception</module>
18         <module>platform-openapi</module>
19     </modules>
20
21     <properties>
22         <java.version>25</java.version>
23         <spring.boot.version>4.0.0</spring.boot.version>
24         <spring.cloud.version>2025.1.0</spring.cloud.version>
25         <springdoc.version>3.0.0</springdoc.version>
26         <mapstruct.version>1.6.3</mapstruct.version>
27         <lombok.version>1.18.42</lombok.version>
28         <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
29         <project.reporting.outputEncoding>UTF-8</project.reporting.outputEncoding>
30     </properties>
31
32     <dependencyManagement>
33         <dependencies>
```

- យើងដាក់ package វាជា pom ដោយសារតែយើងបង្កើតវាជា parent project ហើយបានធ្វើការ add dependency platform យើងមួយៗចូល (it is the base file)

- កន្លែងដែលយើងអាចដាក់ Module យើងចូល

```
<modules>
  <module>platform-common</module>
</modules>
```

- យើងអាចធ្វើការ define កំណត់ទៅលើ Version ផ្សេងៗ បាននៅក្នុង Pom project នឹង ដើម្បី អោយ Children វាចឹងដែល easy to control

```
<properties>
  <java.version>25</java.version>
  <spring.boot.version>4.0.0</spring.boot.version>
  <spring.cloud.version>2025.1.0</spring.cloud.version>
  <springdoc.version>3.0.0</springdoc.version>
  <mapstruct.version>1.6.3</mapstruct.version>
  <lombok.version>1.18.42</lombok.version>
  <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  <project.reporting.outputEncoding>UTF-8</project.reporting.outputEncoding>
</properties>
```

- កន្លែង Dependency management សម្រាប់កំណត់អោយ Child ប្រើប្រាស់ Version ឬ ដើម្បីអោយ sub module ប្រើ version នឹង

```
<dependencyManagement>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-dependencies</artifactId>
      <version>${spring.boot.version}</version>
      <type>pom</type>
      <scope>import</scope>
    </dependency>
    <dependency>
      <groupId>org.springframework.cloud</groupId>
      <artifactId>spring-cloud-dependencies</artifactId>
      <version>${spring.cloud.version}</version>
      <type>pom</type>
      <scope>import</scope>
    </dependency>
  </dependencies>
</dependencyManagement>
```

- Plugin សម្រាប់ add feature ទៅតាម decency នីមួយៗ និង

```
<build>
  <pluginManagement>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <configuration>
          <release>${java.version}</release>
          <parameters>true</parameters>
          <annotationProcessorPaths>
            <path>
              <groupId>org.projectlombok</groupId>
              <artifactId>lombok</artifactId>
              <version>${lombok.version}</version>
            </path>
            <path>
              <groupId>org.mapstruct</groupId>
              <artifactId>mapstruct-processor</artifactId>
              <version>${mapstruct.version}</version>
            </path>
            <path>
              <groupId>org.springframework.boot</groupId>
              <artifactId>spring-boot-configuration-processor</artifactId>
              <version>${spring.boot.version}</version>
            </path>
          </annotationProcessorPaths>
        </configuration>
      </plugin>
    </plugins>
  </pluginManagement>
</build>
```

Code:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.pisethjavaschool.platform</groupId>
  <artifactId>platform-libraries</artifactId>
  <version>1.0.0</version>
  <packaging>pom</packaging>

  <name>platform-libraries</name>
  <description>Reusable platform libraries for the Restaurant KTV enterprise course.</description>

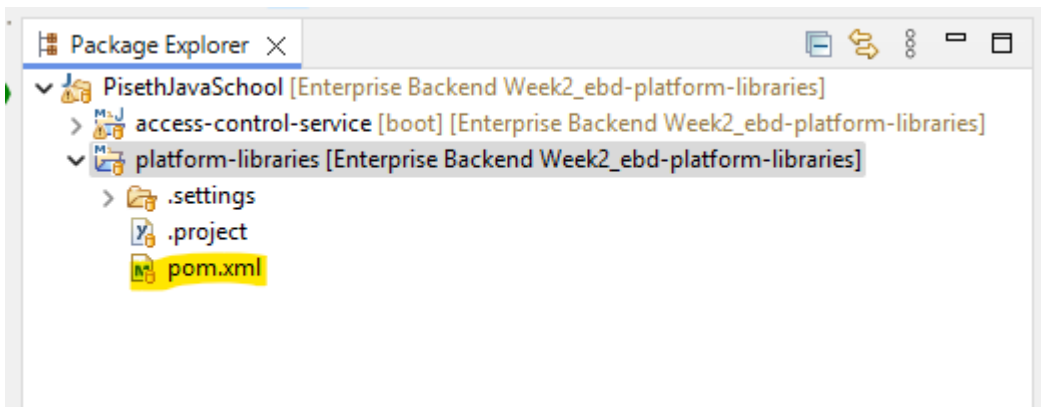
  <modules>
    <module>platform-common</module>
  </modules>

  <properties>
    <java.version>25</java.version>
    <spring.boot.version>4.0.0</spring.boot.version>
    <spring.cloud.version>2025.1.0</spring.cloud.version>
    <springdoc.version>3.0.0</springdoc.version>
    <mapstruct.version>1.6.3</mapstruct.version>
    <lombok.version>1.18.42</lombok.version>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <project.reporting.outputEncoding>UTF-8</project.reporting.outputEncoding>
  </properties>

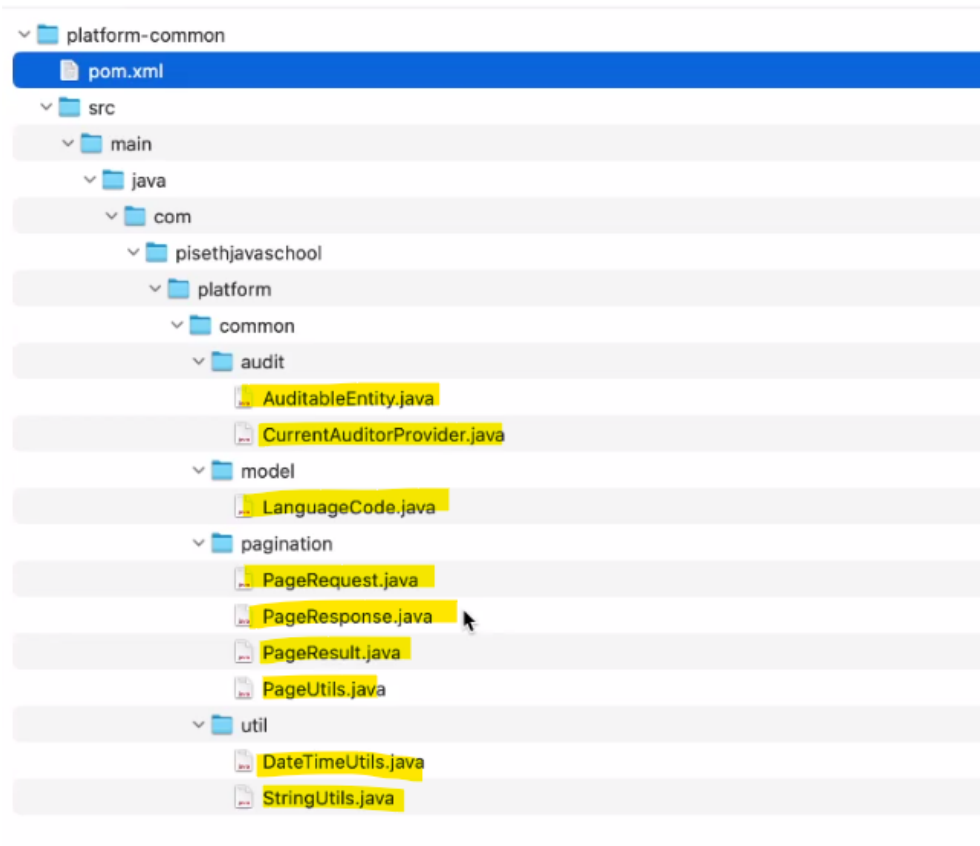
  <dependencyManagement>
    <dependencies>
      <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-dependencies</artifactId>
        <version>${spring.boot.version}</version>
        <type>pom</type>
        <scope>import</scope>
      </dependency>
      <dependency>
        <groupId>org.springframework.cloud</groupId>
        <artifactId>spring-cloud-dependencies</artifactId>
        <version>${spring.cloud.version}</version>
        <type>pom</type>
        <scope>import</scope>
      </dependency>
    </dependencies>
  </dependencyManagement>

  <build>
    <pluginManagement>
      <plugins>
        <plugin>
          <groupId>org.apache.maven.plugins</groupId>
          <artifactId>maven-compiler-plugin</artifactId>
          <configuration>
            <release>${java.version}</release>
            <parameters>true</parameters>
            <annotationProcessorPaths>
              <path>
                <groupId>org.projectlombok</groupId>
                <artifactId>lombok</artifactId>
                <version>${lombok.version}</version>
              </path>
              <path>
                <groupId>org.mapstruct</groupId>
                <artifactId>mapstruct-processor</artifactId>
                <version>${mapstruct.version}</version>
              </path>
              <path>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-configuration-processor</artifactId>
                <version>${spring.boot.version}</version>
              </path>
            </annotationProcessorPaths>
          </configuration>
        </plugin>
      </plugins>
    </pluginManagement>
  </build>
</project>
```

⇒ We can import them to our IDE (Import pom.xml)



2- We build the platform command



Softcode > ebd-platform-libraries

Name	Date modified	Type	Size
.settings	7/27/2026 3:03 PM	File folder	
platform-common	7/27/2026 3:14 PM	File folder	
.project	7/27/2026 3:03 PM	PROJECT File	1 KB
pom.xml	7/27/2026 2:04 PM	XML File	4 KB

2

1 ↗ Paren pom

- យើងយក platform-common មកដាក់ចូលក្នុង Folder “pom” parent (សំខាន់យើងមើលឈ្មោះរបស់ វាអោយត្រូវទៅនឹងអ្វីដែលយើង ចង់ដាក់ចូលនឹង)

```

pom.xml
File Edit View

<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>com.pisethjavaschool.platform</groupId>
    <artifactId>platform-libraries</artifactId>
    <version>1.0.0</version>
  </parent>

  <artifactId>platform-common</artifactId>

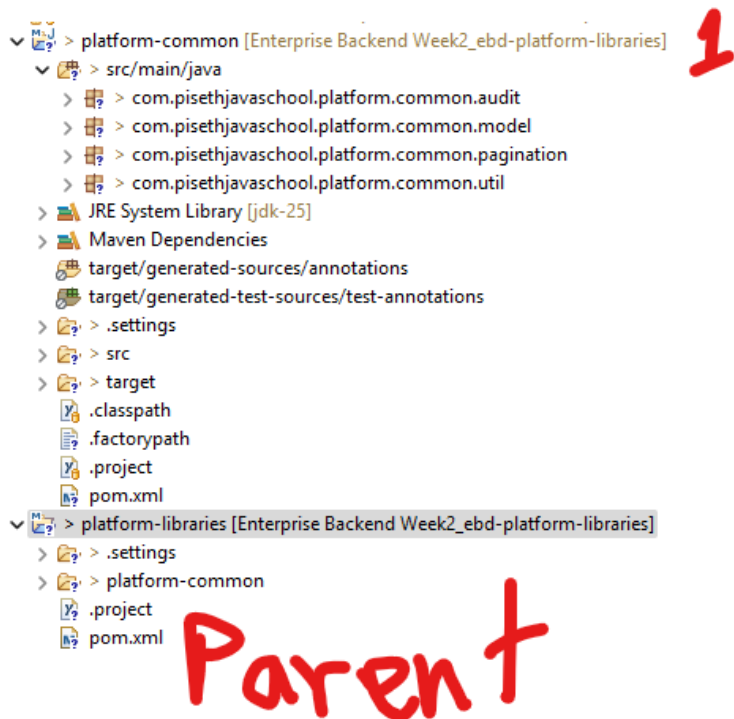
  <packaging>jar</packaging>

  <dependencies>

```

- ហើយយើងត្រូវដាក់ packaging វាជា jar file ព្រោះយើងយកវាមកប្រើ

ចឹងយើងត្រូវធ្វើការ Import វាជា Project ម្តងទៀតទើបយើងអាចធ្វើការ create jar file វាបាន



- inside the pom.xml for the palatform-common :

```

1<project xmlns="http://maven.apache.org/POM/4.0.0"
2  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4                      https://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6  <modelVersion>4.0.0</modelVersion>
7
8  <parent>
9    <groupId>com.pisethjaschool.platform</groupId>
10   <artifactId>platform-libraries</artifactId>
11   <version>1.0.0</version>
12 </parent>
13
14 <artifactId>platform-common</artifactId>
15
16 <packaging>jar</packaging>
17
18 <dependencies>
19
20   <dependency>
21     <groupId>jakarta.validation</groupId>
22     <artifactId>jakarta.validation-api</artifactId>
23   </dependency>
24
25 </dependencies>

```

2 Link 2 P

Build the Library from the parent file

```

> platform-libraries [Enterprise Backend Week2_ebd-platform-libraries]
  > .settings
  > platform-common
    .project
    pom.xml

```

>> mvn clean install

We use the mvn clean install ដើម្បីធ្វើការ បង្កើត Jar file ហើយដាក់វាចូលទៅក្នុង Local repository .m2

```

[INFO] --- install:3.1.2:install (default-install) @ platform-common ---
[INFO] Installing C:\Users\home\Desktop\Enterprise Backend\Softcode\ebd-platform-libra
1 to C:\Users\home\.m2\repository\com\pisethjaschool\platform\platform-common\1.0.0\
[INFO] Installing C:\Users\home\Desktop\Enterprise Backend\Softcode\ebd-platform-libra
\platform-common-1.0.0.jar to C:\Users\home\.m2\repository\com\pisethjaschool\platf
atform-common-1.0.0.jar
[INFO] -----
[INFO] Reactor Summary for platform-libraries 1.0.0:
[INFO]
[INFO] platform-libraries ..... SUCCESS [ 0.256 s]
[INFO] platform-common ..... SUCCESS [ 2.305 s]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.816 s
[INFO] Finished at: 2026-07-27T15:45:56+07:00
[INFO] -----

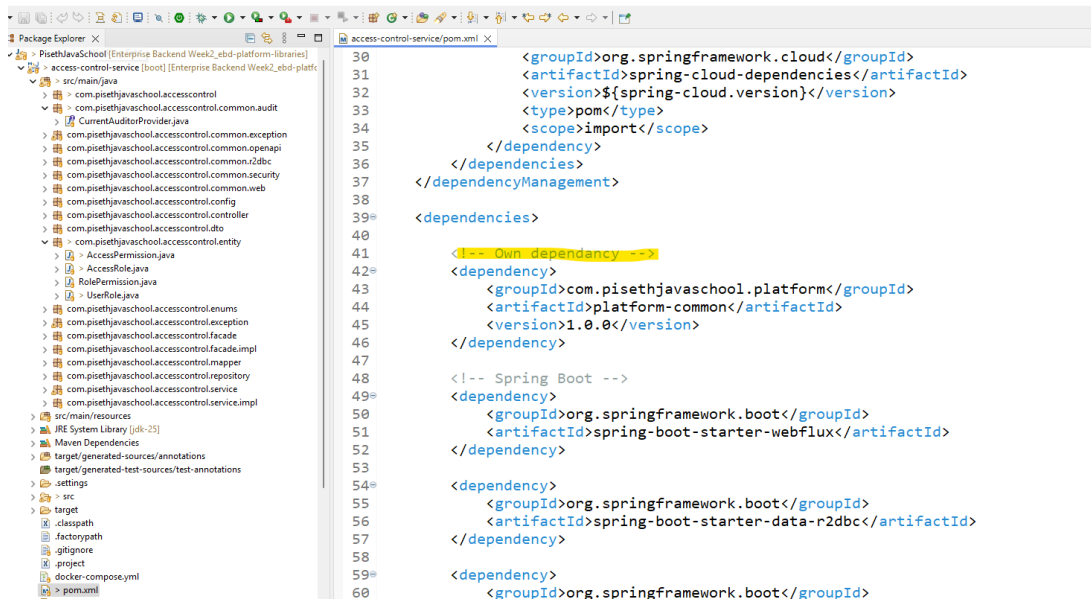
```

.m2 > repository > com > pisethjaschool > platform > platform-common > 1.0.0

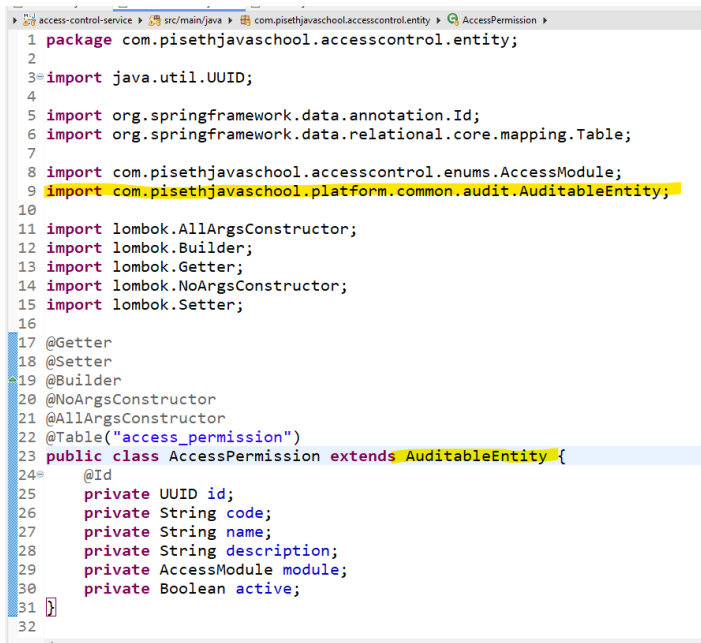
Name	Date modified
_remote.repositories	7/27/2026 3:45 PM
platform-common-1.0.0.jar	7/27/2026 3:45 PM
platform-common-1.0.0.pom	7/26/2026 2:43 PM

- Import the jar file to the project

Add the dependance:



```
<dependency>
    <groupId>com.pisethjaschool.platform</groupId>
    <artifactId>platform-common</artifactId>
    <version>1.0.0</version>
</dependency>
```

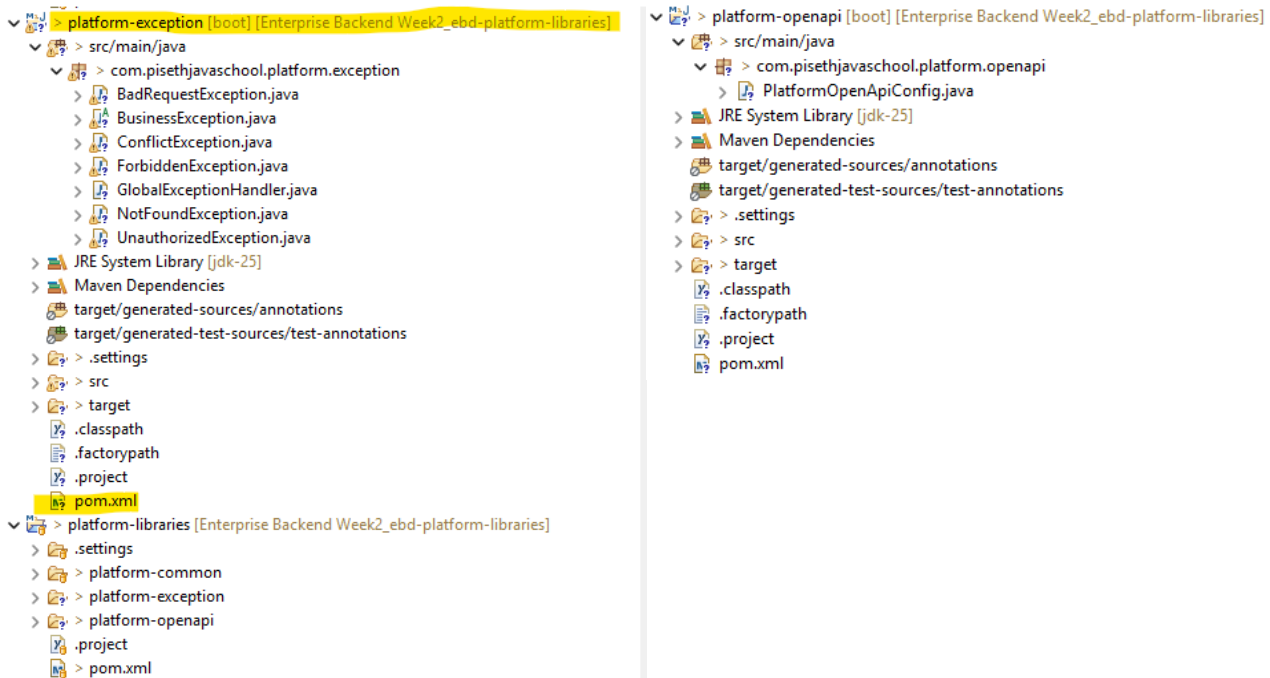


- In the class we can import teams from our package

Import Exception

1- Add platform exception

- Import exception to STS: ដើម្បីធ្វើការ compiler creates jar file to the .me (our local repository). And Import openapi to STS.



- In the pom.xml (parent): ដើម្បីអោយវាដាក់ dependency from parent class

```
<modules>
  <module>platform-common</module>
  <module>platform-exception</module>
  <module>platform-openapi</module>
</modules>
```

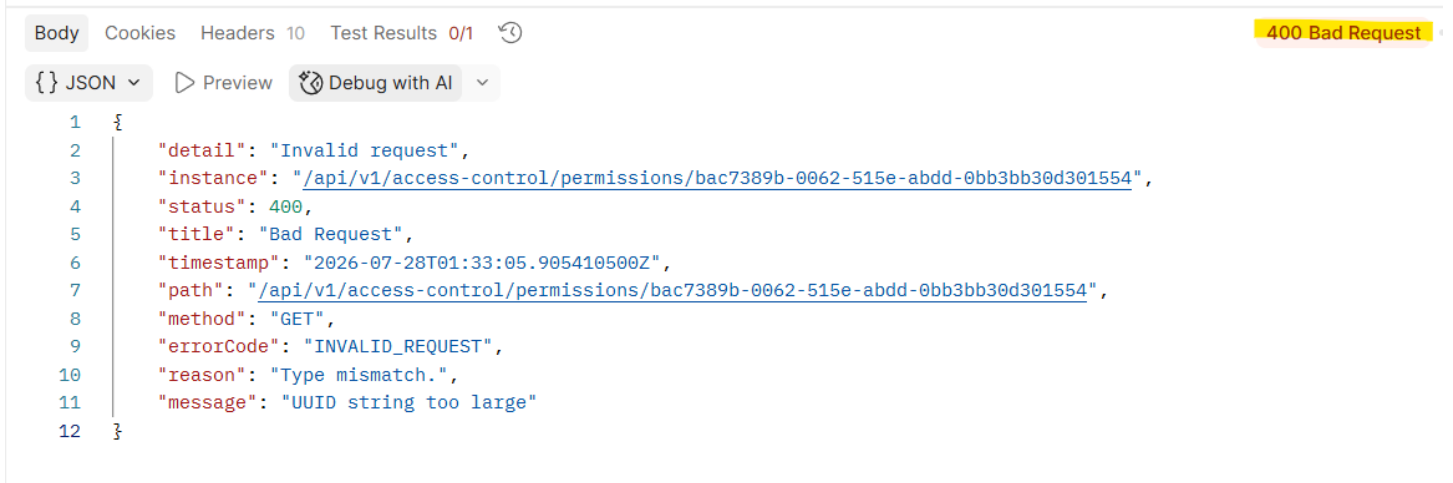
⇒ Where we want to use the library, we can add the dependency

```
<!-- Own dependency -->
<dependency>
  <groupId>com.pisethjavaschool.platform</groupId>
  <artifactId>platform-common</artifactId>
  <version>1.0.0</version>
</dependency>

<dependency>
  <groupId>com.pisethjavaschool.platform</groupId>
  <artifactId>platform-exception</artifactId>
  <version>1.0.0</version>
</dependency>

<dependency>
  <groupId>com.pisethjavaschool.platform</groupId>
  <artifactId>platform-openapi</artifactId>
  <version>1.0.0</version>
</dependency>
<!-- End of dependency -->
```

- Testing api class



Body Cookies Headers 10 Test Results 0/1 400 Bad Request

{ } JSON Preview Debug with AI

```

1 {
2   "detail": "Invalid request",
3   "instance": "/api/v1/access-control/permissions/bac7389b-0062-515e-abdd-0bb3bb30d301554",
4   "status": 400,
5   "title": "Bad Request",
6   "timestamp": "2026-07-28T01:33:05.905410500Z",
7   "path": "/api/v1/access-control/permissions/bac7389b-0062-515e-abdd-0bb3bb30d301554",
8   "method": "GET",
9   "errorCode": "INVALID_REQUEST",
10  "reason": "Type mismatch.",
11  "message": "UUID string too large"
12 }

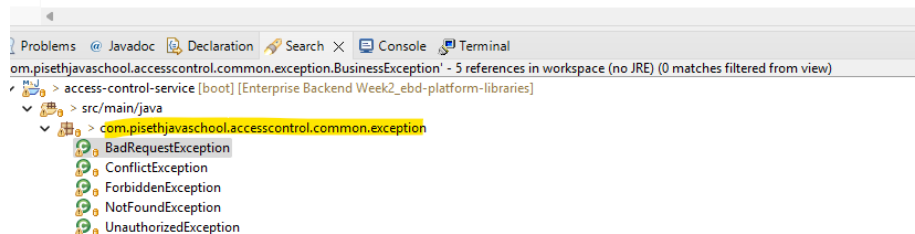
```

- We can check the implementation from the getByld

```

1 package com.pisethjavaschool.accesscontrol.common.exception;
2
3 import org.springframework.http.HttpStatus;
4
5 public class NotFoundException extends BusinessException {
6     public NotFoundException(String message) {
7         super("RESOURCE_NOT_FOUND", message, HttpStatus.NOT_FOUND);
8     }
9
10    public NotFoundException(String errorCode, String message) {
11        super(errorCode, message, HttpStatus.NOT_FOUND);
12    }
13 }
14

```



Problems Javadoc Declaration Search Console Terminal

com.pisethjavaschool.accesscontrol.common.exception.BusinessException' - 5 references in workspace (no JRE) (0 matches filtered from view)

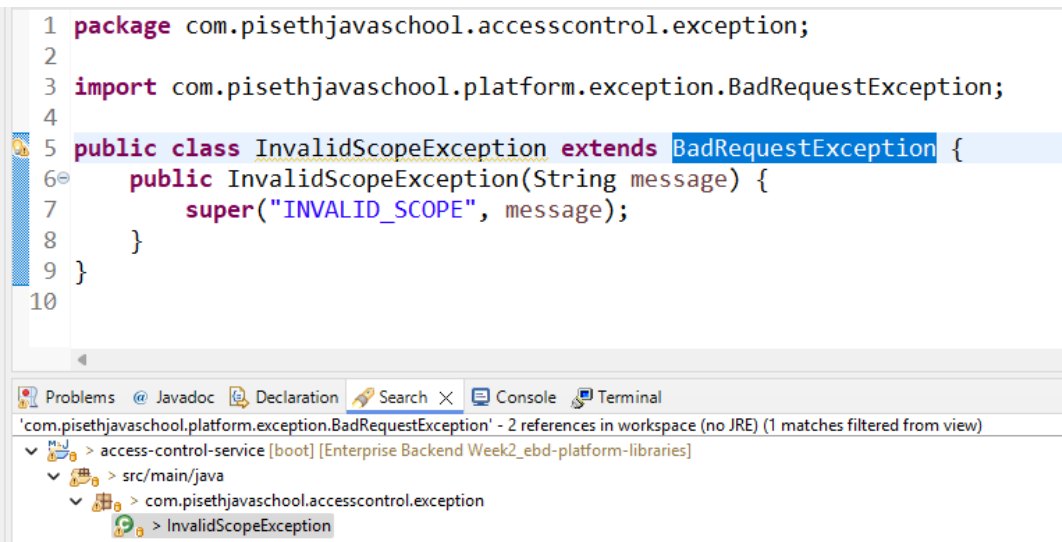
> access-control-service [boot] [Enterprise Backend Week2_ebd-platform-libraries]

> src/main/java

> com.pisethjavaschool.accesscontrol.common.exception

- BadRequestException
- ConflictException
- ForbiddenException
- NotFoundException
- UnauthorizedException

កន្លែងនឹងមានន័យថា វាចាប់យក Package from their local package → so we can deleted it the whold exception from the class. យើងប្តូរ Package វាជាក់ទៅកាន់ Pacakge platform យើងវិញ។



```

1 package com.pisethjavaschool.accesscontrol.exception;
2
3 import com.pisethjavaschool.platform.exception.BadRequestException;
4
5 public class InvalidScopeException extends BadRequestException {
6     public InvalidScopeException(String message) {
7         super("INVALID_SCOPE", message);
8     }
9 }
10

```

Problems Javadoc Declaration Search Console Terminal

'com.pisethjavaschool.platform.exception.BadRequestException' - 2 references in workspace (no JRE) (1 matches filtered from view)

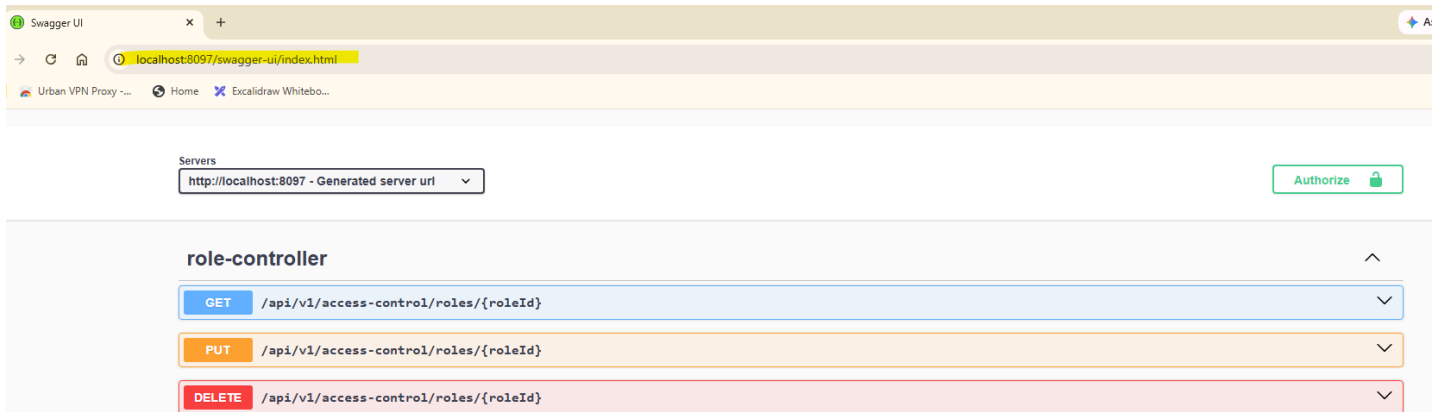
> access-control-service [boot] [Enterprise Backend Week2_ebd-platform-libraries]

> src/main/java

> com.pisethjavaschool.accesscontrol.exception

- InvalidScopeException

- Replace swagger from our platform



- We just add the dependencies from the platform to our hery

```
<!-- Own dependency -->
<dependency>
  <groupId>com.pisethjavaschool.platform</groupId>
  <artifactId>platform-common</artifactId>
  <version>1.0.0</version>
</dependency>

<dependency>
  <groupId>com.pisethjavaschool.platform</groupId>
  <artifactId>platform-exception</artifactId>
  <version>1.0.0</version>
</dependency>

<dependency>
  <groupId>com.pisethjavaschool.platform</groupId>
  <artifactId>platform-openapi</artifactId>
  <version>1.0.0</version>
</dependency>
<!-- End of dependency -->
```

⇒ Then it will work for the openapi swagger